DDIN TC Agent Visit Reporting System

Technical Implementation Documentation

Version 1.0

1. Executive Summary

1.1 Document Purpose

This document provides comprehensive technical specifications for the implementation of the DDIN TC Agent Visit Reporting System. The system transforms paper-based field visits into a fully digital, verifiable process using a Progressive Web Application (PWA) architecture.

1.2 System Overview

Territory Coordinators (TCs) visit DDIN agents in person, and every visit is recorded in the digital system. Each visit record maintains a permanent link between the TC, the agent, the agent's registered location, the TC's real GPS location, the date and time, a written report, and photographic evidence.

1.3 Key Requirements

Requirement	Description
GPS Verification	TC must be physically present at agent's location
Digital Report	Structured visit report with configurable checklist
Photo Evidence	Mandatory photo captured during every visit
Offline Capability	Full functionality without internet connection
Historical Tracking	All assignment and location changes are preserved

2. System Architecture

2.1 High-Level Architecture Diagram

text

CLIENT LAYER

Progressive Web Application (PWA)

Responsive Design (Mobile-first)

Service Workers for Offline Support

Web App Manifest

Push Notifications

GPS/Geolocation API

Camera API Integration

IndexedDB for Local Storage

API GATEWAY

Authentication/Authorization (JWT)

Rate Limiting

Request Validation

SSL Termination

WebSocket for Real-time Updates

APPLICATION LAYER

Microservices:

Auth Service

Visits Service

Agents Service

Reports Service

GPS Service

Photo Service

Analytics Service

DATA LAYER

PostgreSQL

Redis Cache

Object Storage

PostGIS

Session

Photos

Main DB

Rate Limits

Reports

Geo Queries

Temp Data

Documents

Elasticsearch (Search & Analytics)

2.2 Architecture Principles

Principle	Description
PWA-First	Mobile-first Progressive Web Application for all devices
Offline-Ready	Full functionality available without internet connection
Security-First	Encryption, authentication, and authorization at every layer
Scalability	Microservices architecture supports growth
Observability	Comprehensive monitoring and logging

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3. Technology Stack

3.1 Frontend - Progressive Web Application

Component	Technology	Justification
Framework	React 18+ / Vue 3+	Modern SPA, excellent PWA support
PWA Core	Workbox + Service Workers	Offline support, caching strategies
State Management	Redux Toolkit / Zustand	Predictable state management
UI Framework	Material-UI / Tailwind	Production-ready components
Maps	Mapbox GL JS / Leaflet	Interactive mapping, GPS visualization
Camera	WebRTC / MediaDevices API	Native camera access from browser
Geolocation	Geolocation API / Navigator	GPS capture, continuous tracking

Component	Technology	Justification
Storage	IndexedDB + localStorage	Offline data persistence
Push	Push API + Notification API	Real-time alerts

3.2 Backend Technologies

Component	Technology	Justification
API Framework	Node.js + Express / Python FastAPI	High performance, microservices-ready
API Gateway	Kong / Traefik	Rate limiting, routing, observability
Main Database	PostgreSQL 15+ with PostGIS	GIS capabilities, ACID compliance
Cache	Redis 7+	Session management, rate limiting
Search	Elasticsearch 8+	Full-text search, analytics
Message Queue	RabbitMQ / Apache Kafka	Async processing, reliability
Object Storage	AWS S3 / MinIO	Scalable photo/document storage
Container	Docker + Kubernetes	Scalability, orchestration

3.3 Infrastructure

Component	Technology	Justification
Cloud Platform	AWS / Azure / GCP	Production-ready, scalable
Database Hosting	AWS RDS / Azure Database	Managed PostgreSQL, backups

Component	Technology	Justification
Container Registry	AWS ECR / Docker Hub	Secure image storage
Orchestration	AWS EKS / GKE	Kubernetes-managed scaling
CDN	CloudFront / Cloudflare	Fast photo delivery

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4. PWA Implementation Strategy

4.1 Service Worker Architecture

SERVICE WORKER LIFECYCLE

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1. **Installation:**

Cache static assets (HTML, CSS, JS)

Cache critical API responses (agents list, etc.)

Initialize IndexedDB schema

2. **Activation:**

Clear old caches

Claim clients

Register background sync

3. **Fetch Handling:**

Network first for dynamic data

Cache first for static assets

Stale-while-revalidate for agents data

Cache API responses for offline mode

Background Sync:

Queue offline visits

Retry failed photo uploads

Sync reports when online

4.2 Offline Storage Strategy

OFFLINE STORAGE

IndexedDB - Primary Offline Store:

Object Stores:

visits (offline visit records)

photos (blob data of captured photos)

agents (cached agent data

sync_queue (pending sync operations)

audit_logs (offline activity logs)

Cache Storage (Service Worker):

Static Assets (app shell)

API Responses (GET requests)

Images (thumbnails, logos)

LocalStorage:

User preferences

Last known location

Session tokens

4.3 PWA Manifest Configuration

Property	Value	Description
Name	DDIN TC Visit System	Full application name
Short Name	DDIN Visits	Short display name
Description	TC Agent Visit Reporting System	Application description
Theme Color	#1976D2	Brand color for browser UI
Background Color	#FFFFFF	Splash screen background
Display	standalone	Full-screen app experience
Orientation	portrait	Mobile-first orientation
Start URL	/dashboard	Home page after launch
Scope	/	Full application scope

4.4 App Icons Required

Size	Purpose
72x72	Android launcher (mdpi)
96x96	Android launcher (hdpi)
144x144	Android launcher (xhdpi)
192x192	Android launcher (xxhdpi)

Size	Purpose
512x512	Android launcher (xxxhdpi)
180x180	Apple touch icon

5. Database Schema Design

5.1 Core Tables

Users Table

Stores all system users including TCs, Champions, Managers, and Admins.

Field	Type	Description
id	UUID	Primary key
username	VARCHAR(100)	Unique username
email	VARCHAR(255)	Unique email address
password_hash	VARCHAR(255)	Bcrypt hashed password
role	VARCHAR(50)	admin, manager, tc, champion, agent
first_name	VARCHAR(100)	User's first name
last_name	VARCHAR(100)	User's last name
phone_number	VARCHAR(20)	Contact phone number
is_active	BOOLEAN	Account status

Field	Type	Description
last_login	TIMESTAMP	Last login timestamp

Territories Table

Manages organizational structure (Districts and Sectors).

Field	Type	Description
id	UUID	Primary key
name	VARCHAR(200)	Territory name
type	VARCHAR(50)	district or sector
parent_id	UUID	Reference to parent territory
code	VARCHAR(50)	Unique territory code
is_active	BOOLEAN	Active status

TC Territory Assignments

Supports TC coverage of multiple districts.

Field	Type	Description
id	UUID	Primary key
tc_id	UUID	Reference to TC user
territory_id	UUID	Reference to territory
assigned_at	TIMESTAMP	Assignment start date

Field	Type	Description
assigned_by	UUID	Who made the assignment
end_date	TIMESTAMP	Assignment end date
is_active	BOOLEAN	Currently active assignment

Agents Table

Stores all agent information including GPS location.

Field	Type	Description
id	UUID	Primary key
agent_id	VARCHAR(50)	Unique agent identifier
name	VARCHAR(200)	Agent's full name
phone_number	VARCHAR(20)	Agent's phone number
business_name	VARCHAR(200)	Business name
address	TEXT	Physical address
territory_id	UUID	Reference to district
sector_id	UUID	Reference to sector
champion_id	UUID	Assigned champion
status	VARCHAR(50)	active, inactive, suspended, pending

Field	Type	Description
registered_latitude	DECIMAL(10,8)	GPS latitude
registered_longitude	DECIMAL(11,8)	GPS longitude
gps_accuracy	DECIMAL(10,2)	GPS accuracy in meters
location_verified	BOOLEAN	Location verification status

Visits Table

Core visit record with all verification data.

Field	Type	Description
id	UUID	Primary key
visit_id	VARCHAR(50)	Unique visit identifier
tc_id	UUID	Reference to TC user
agent_id	UUID	Reference to agent
territory_id	UUID	Territory at visit time
visit_date	DATE	Visit date
check_in_time	TIMESTAMP	Visit start time
check_out_time	TIMESTAMP	Visit end time
agent_latitude	DECIMAL(10,8)	Agent's registered location

Field	Type	Description
agent_longitude	DECIMAL(11,8)	Agent's registered location
tc_latitude	DECIMAL(10,8)	TC's actual GPS location
tc_longitude	DECIMAL(11,8)	TC's actual GPS location
distance_meters	DECIMAL(10,2)	Calculated distance
gps_accuracy_meters	DECIMAL(10,2)	GPS reading accuracy
location_status	VARCHAR(50)	verified, outside_radius, low_accuracy
photo_id	UUID	Reference to visit photo
visit_status	VARCHAR(50)	started, completed, flagged, rejected
is_offline	BOOLEAN	Record created offline
sync_status	VARCHAR(50)	Sync status

Visit Reports Table

Structured visit report data.

Field	Type	Description
id	UUID	Primary key
visit_id	UUID	Reference to visit
operating_status	VARCHAR(50)	operational, temporarily_closed, etc.

Field	Type	Description
services_offered	TEXT	Services provided by agent
products_offered	TEXT	Products offered
general_observations	TEXT	General activity observations
liquidity_observations	TEXT	Cash-float observations
customer_quality_observations	TEXT	Service quality observations
challenges_faced	TEXT	Challenges the agent is facing
support_needs	TEXT	Support requirements
training_needs	TEXT	Training requirements
follow_up_actions	TEXT	Required follow-up actions
additional_remarks	TEXT	Additional TC remarks
checklist_data	JSONB	Configurable checklist answers

Visit Photos Table

Stores visit photo metadata.

Field	Type	Description
id	UUID	Primary key
visit_id	UUID	Reference to visit

Field	Type	Description
photo_url	VARCHAR(500)	Cloud storage URL
thumbnail_url	VARCHAR(500)	Thumbnail URL
photo_captured_at	TIMESTAMP	Capture timestamp
file_size_bytes	BIGINT	Photo file size
mime_type	VARCHAR(100)	Image format
original_filename	VARCHAR(255)	Original file name

Audit Logs Table

Tracks all system actions for compliance.

Field	Type	Description
id	UUID	Primary key
user_id	UUID	User who performed action
action	VARCHAR(100)	Action type
resource_type	VARCHAR(50)	Type of resource affected
resource_id	UUID	Resource identifier
details	JSONB	Action details
ip_address	INET	User's IP address

Field	Type	Description
user_agent	TEXT	Browser/device information

5.2 PostGIS Spatial Functions

The system uses PostGIS for all location-based calculations:

1. Calculate Distance: Function to calculate distance between TC and agent coordinates
2. Find Agents in Radius: Function to find agents within a specified radius
3. Location Validation: Function to validate TC location against agent location

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6. API Design

6.1 Authentication API

Endpoint	Method	Description
/api/v1/auth/login	POST	User login with credentials
/api/v1/auth/refresh	POST	Refresh access token
/api/v1/auth/logout	POST	User logout
/api/v1/auth/forgot-password	POST	Request password reset
/api/v1/auth/reset-password	POST	Reset password with token

Login Request

text

```
{
  "username": "tc023",
  "password": "secure_password",
  "device_info": {
```

```
    "device_id": "device_123",
    "platform": "mobile",
    "version": "1.0.0"
  }
}
```

Login Response

text

```
{
  "access_token": "jwt_token_here",
  "refresh_token": "refresh_token_here",
  "expires_in": 3600,
  "user": {
    "id": "uuid",
    "username": "tc023",
    "role": "tc",
    "name": "John Doe",
    "permissions": ["agents.view", "visits.create"],
    "assigned_territories": ["district_001", "district_002"]
  }
}
```

6.2 Agent Management API

Endpoint	Method	Description
/api/v1/agents	GET	List agents with filters
/api/v1/agents	POST	Create new agent

Endpoint	Method	Description
/api/v1/agents/:id	GET	Get agent details
/api/v1/agents/:id	PUT	Update agent information
/api/v1/agents/:id/location	PATCH	Update agent location
/api/v1/agents/:id/location-history	GET	Get location history

Agent Creation Request

text

```
{
  "agent_id": "AGT-00245",
  "name": "Jane Doe",
  "phone_number": "+250788123456",
  "business_name": "Doe Enterprises",
  "address": "Kigali, Rwanda",
  "territory_id": "dist_001",
  "sector_id": "sect_001",
  "latitude": -1.9441,
  "longitude": 30.0619,
  "gps_accuracy": 15.5
}
```

6.3 Visit Management API

Endpoint	Method	Description
/api/v1/visits/check-in	POST	Start a visit

Endpoint	Method	Description
/api/v1/visits/:id/photo	PUT	Upload visit photo
/api/v1/visits/:id/report	POST	Submit visit report
/api/v1/visits/:id/check-out	PUT	Complete the visit
/api/v1/visits	GET	List visits with filters
/api/v1/visits/:id	GET	Get visit details

Check-in Request

text

```
{
  "agent_id": "AGT-00245",
  "latitude": -1.9440,
  "longitude": 30.0618,
  "gps_accuracy": 12.3
}
```

Check-in Response

text

```
{
  "visit_id": "VIS-2024-001",
  "location_status": "verified",
  "distance_meters": 32.5,
  "allowed_radius": 100,
  "timestamp": "2024-01-15T09:30:00Z"
}
```

Visit Report Request

text


```
{
  "operating_status": "operational",
  "services_offered": "Mobile money, Airtime, Bill payments",
  "products_offered": "MTN, Tigo, Airtel",
  "general_observations": "Agent is busy with customers",
  "liquidity_observations": "Sufficient cash available",
  "customer_quality_observations": "Good customer handling",
  "challenges_faced": "Network issues occasionally",
  "support_needs": "Need more marketing materials",
  "training_needs": "Advanced mobile money training",
  "follow_up_actions": "Send marketing materials next week",
  "additional_remarks": "Agent requested more visibility",
  "checklist_data": {
    "question_1": "yes",
    "question_2": "no",
    "question_3": "partial"
  }
}
```


6.4 GPS Validation API

Endpoint	Method	Description
/api/v1/gps/validate	POST	Validate GPS location
/api/v1/gps/geofence-config	GET	Get geofence configuration

Endpoint	Method	Description
/api/v1/gps/geofence-config	PUT	Update geofence radius

GPS Validation Request

text


```
{
  "tc_latitude": -1.9440,
  "tc_longitude": 30.0618,
  "tc_accuracy": 12.3,
  "agent_id": "AGT-00245"
}
```


GPS Validation Response

text


```
{
  "is_valid": true,
  "distance_meters": 32.5,
  "allowed_radius": 100,
  "location_status": "verified",
  "details": {
    "tc_coordinates": {
      "latitude": -1.9440,
      "longitude": 30.0618
    },
    "agent_coordinates": {
      "latitude": -1.9441,
      "longitude": 30.0619
    },
```

```
    "accuracy": 12.3,
    "timestamp": "2024-01-15T09:30:00Z"
  }
}
```


6.5 Reporting API

Endpoint	Method	Description
/api/v1/reports/visits	GET	Generate visit report
/api/v1/reports/coverage	GET	Generate coverage report
/api/v1/reports/analytics/dashboard	GET	Dashboard analytics
/api/v1/reports/export	GET	Export reports (CSV/PDF)
/api/v1/reports/audit-logs	GET	View audit logs

Visit Report Query Parameters

text

GET /api/v1/reports/visits?from_date=2024-01-01&to_date=2024-01-31&tc_id=tc_001&format=json

6.6 Configuration API

Endpoint	Method	Description
/api/v1/config/checklist	GET	Get visit checklist
/api/v1/config/checklist	PUT	Update visit checklist
/api/v1/config/geofence	GET	Get geofence config

Endpoint	Method	Description
/api/v1/config/system	GET	Get system config

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7. Security Implementation

7.1 Authentication & Authorization

JWT Token Structure

text

Header:

- alg: HS256/RS256
- typ: JWT

Payload:

- sub: user_id
- username: string
- role: string
- permissions: string[]
- territories: string[]
- exp: timestamp
- iat: timestamp
- iss: issuer
- jti: token_id

7.2 Role-Based Access Control Matrix

Permission	Admin	Manager	TC	Champion
Manage Users	✓	✗	✗	✗

Permission	Admin	Manager	TC	Champion
Manage Agents	✓	✓	✓	✓
Create Visits	✓	✗	✓	✗
View Reports	✓	✓	✓	✓
Export Data	✓	✓	✗	✗
Configure System	✓	✗	✗	✗
View All Agents	✓	✓	✗	✗
View Sector Agents	✓	✓	✓	✓
View District Agents	✓	✓	✓	✗

7.3 Data Encryption

Data Type	Encryption Method	Key Management
Passwords	Bcrypt (work factor 12)	Application level
PII Data	AES-256-GCM	Customer-managed keys
Photos	Server-side encryption	AWS KMS / Azure Key Vault
API Traffic	TLS 1.3	Managed certificates
Database	Transparent Data Encryption	Cloud provider managed

7.4 Security Headers

Header	Value	Purpose
Content-Security-Policy	strict	Prevent XSS attacks
X-Frame-Options	DENY	Prevent clickjacking
X-Content-Type-Options	nosniff	Prevent MIME sniffing
Referrer-Policy	strict-origin-when-cross-origin	Control referrer info
Permissions-Policy	geolocation, camera	Control device APIs

7.5 Audit Trail

All critical actions are logged in the audit trail:

Action	Resource Type	Logged Data
Login	User	User ID, IP, timestamp
Visit Creation	Visit	TC, Agent, GPS data
Photo Upload	Photo	File size, timestamp
Report Submission	Report	Report data, timestamp
Agent Update	Agent	Changed fields, user
Configuration Change	Config	Old value, new value, user
User Management	User	Action type, user

8. Offline Capability

8.1 Offline Data Strategy

The PWA supports full offline functionality with the following strategy:

1. Data Caching: Critical data (agents, configurations) is cached on first load
2. Offline Operations: Visits can be created, photos captured, and reports written offline
3. Local Storage: All offline data stored in IndexedDB with structured stores
4. Background Sync: Automatic synchronization when network is available
5. Conflict Resolution: Server wins on ID conflicts, timestamps for ordering

8.2 Offline Storage Limits

Storage Type	Limit	Cleanup Policy
IndexedDB	Up to 50MB	Old visits archived
Cache Storage	Up to 100MB	LRU cache eviction
Photo Storage	Last 50 visits	Oldest photos removed
Sync Queue	Unlimited	Processed items removed

8.3 Background Sync Triggers

Trigger	Action
Application Opens	Auto-sync pending data
Network Reconnects	Background sync started
Periodic (15 min)	Background sync while online
Push Notification	Manual sync triggered
User Initiated	Manual sync with progress UI

9. Mobile-First Design

9.1 Responsive Breakpoints

Device Type	Screen Width	Layout
Mobile	< 600px	Single column, bottom navigation
Tablet	600px - 1024px	Two-column, side navigation
Desktop	> 1024px	Full dashboard, multi-column

9.2 Touch Optimization Guidelines

- Minimum touch target: 44×44px
- Touch feedback with ripple effects
- Gesture support: pull-to-refresh, swipe navigation
- Form inputs sized for finger tapping
- Buttons: minimum 48px height
- Adequate spacing between interactive elements
- Focus indicators and ARIA labels for accessibility

9.3 Camera Integration

The PWA uses the MediaDevices API for camera access:

1. Open Camera: Direct access to device camera
2. Capture Photo: High-resolution photo capture
3. Local Compression: Compress and optimize locally
4. Offline Storage: Store in IndexedDB with metadata
5. GPS Embed: Embed GPS coordinates in EXIF data
6. Thumbnail Generation: Create preview thumbnails

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10. Monitoring & Observability

10.1 Key Performance Metrics

Metric	Target	Monitoring Tool
First Contentful Paint	< 1.5s	Lighthouse/Core Web Vitals
Time to Interactive	< 3.0s	Lighthouse/Core Web Vitals
Largest Contentful Paint	< 2.5s	Lighthouse/Core Web Vitals
Cumulative Layout Shift	< 0.1	Lighthouse/Core Web Vitals
API Response Time	< 500ms (p95)	APM (Datadog/New Relic)
Service Worker Activation	< 2s	Custom metrics
IndexedDB Operations	< 100ms	Custom metrics
GPS Acquisition Time	< 5s	Custom metrics

10.2 Error Tracking

Error Type	Monitoring Tool	Action
API Failures	Sentry/Logs	Alert, Retry
GPS Failures	Custom Metrics	User Feedback
Sync Failures	Sync Queue	Retry, Alert
Photo Upload Failures	Queue Tracking	Retry, Fallback

Error Type	Monitoring Tool	Action
Service Worker Errors	Console Logs	Rollback
Offline Storage Full	Custom Metrics	Cleanup, Alert

10.3 Logging Strategy

Log Type	Purpose	Retention
Application Logs	Debugging and troubleshooting	30 days
Access Logs	Security and compliance	90 days
Audit Logs	User action tracking	7 years
Performance Logs	Capacity planning	90 days
Error Logs	Issue identification	90 days

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11. Deployment & DevOps

11.1 CI/CD Pipeline

Pipeline Stages:

1. Code Commit: Developer pushes code to repository
2. Unit Tests: Automated unit tests run
3. Build: Application built for production
4. Integration Tests: API integration tests run
5. E2E Tests: End-to-end tests with Cypress/Playwright
6. Docker Build: Container images built
7. Registry Push: Images pushed to container registry

8. Staging Deploy: Deploy to staging environment
9. Smoke Tests: Critical functionality verified
10. Production Deploy: Blue-green deployment to production
11. Post-Deploy Tests: Health checks and smoke tests
12. Monitoring: Metrics and alerts monitored

11.2 Rollback Strategy

Trigger	Action	Timeframe
Failed Health Checks	Automatic rollback	< 2 minutes
Manual Rollback	Via CI/CD pipeline	< 5 minutes
Previous Version	Kept for 24 hours	-
Database Rollback	Database migration reversal	< 10 minutes
Feature Flags	Safe deployment control	Immediate

11.3 Infrastructure as Code

Cloud Infrastructure (Terraform):

Resource	Configuration
VPC	Private subnets for internal services
Kubernetes Cluster	Managed EKS/GKE/AKS
PostgreSQL	Multi-AZ, encrypted, automated backups
Redis	Cluster mode, multi-AZ
S3 Bucket	Versioned, encrypted, lifecycle policies

CloudFront	CDN with SSL certificate
Route53	DNS management

Container Orchestration (Kubernetes):

Component	Configuration
API Service	2-10 replicas, HPA enabled
Redis	StatefulSet, 3 replicas
PostgreSQL	External RDS (managed)
Nginx Ingress	SSL termination, rate limiting
Cert Manager	Automatic SSL certificates
Service Monitor	Prometheus metrics collection

12. Success Metrics

12.1 Technical Success Criteria

Criterion	Target
PWA Installable	On all major browsers
Digital Visit Capture	100% of TC visits
Offline Capability	All core features offline
GPS Verification	Within 100m accuracy
Load Time (3G)	< 3 seconds

API Uptime	99.9%
API Response Time (p95)	< 500ms
Audit Trail	100% of actions logged

12.2 Business Success Criteria

Criterion	Target
Agent GPS Registration	100% of agents
Visit Completion Rate	95%
Monthly Coverage Rate	80%+
Photo Capture Compliance	100%
Reporting	Real-time capability
Historical Data	Preserved, never overwritten
Administrative Overhead	Reduced

13. Key User Flows

13.1 TC Visit Flow

Step	Action	Description	Offline Support
1	Login	TC logs in with credentials	✗ (requires auth)
2	Dashboard	View assigned agents list	✓

3	Select Agent	Choose agent to visit	✓
4	Request Check-in	System captures GPS	✓
5	GPS Verification	Compare with agent location	✓
6	Capture Photo	Take visit photo	✓
7	Complete Report	Answer structured questions	✓
8	Check-out	Submit visit record	✓
9	Sync	Background sync when online	✓

13.2 Admin/Management Flow

Step	Action	Description	Permissions
1	Dashboard	View real-time KPIs	Admin/Manager
2	Agent Management	Register, update, assign agents	Admin/Manager/TC
3	TC Management	Create accounts, assign territories	Admin/Manager
4	Reporting	Generate, filter, export reports	Admin/Manager
5	Configuration	Adjust geofence, checklist	Admin only
6	Audit	View audit logs	Admin only

14. Key Performance Indicators (KPIs)

14.1 Business KPIs

KPI	Formula	Target	Tracking
Visit Coverage	(Agents Visited ÷ Agents Due) × 100	> 80% monthly	Daily/Weekly/Monthly
Visit Completion Rate	(Completed Visits ÷ Started Visits) × 100	> 90%	Daily/Weekly
Location Verification Rate	(Verified Visits ÷ Total Visits) × 100	> 85%	Daily/Weekly
Photo Capture Rate	(Visits with Photo ÷ Total Visits) × 100	100%	Real-time
Offline Usage Rate	(Offline Visits ÷ Total Visits) × 100	Track trends	Monthly

14.2 System KPIs

KPI	Target	Description
API Uptime	99.9%	Service availability
API Response Time	< 500ms	p95 latency
PWA Load Time	< 3s	On 3G connection
Sync Success Rate	> 95%	Offline sync success
Photo Upload Success	> 90%	Photo upload success
GPS Acquisition Success	> 85%	GPS signal acquisition
Concurrent Users	500+	Supported users

15. Data Privacy & Compliance

15.1 Data Classification

Data Type	Classification	Protection Level
PII (Personal)	High	Encryption at rest and in transit
GPS Coordinates	Medium	Access control, audit logging
Photos	High	Secure storage, access control
Visit Reports	Medium	Access control, audit logging
Audit Logs	High	Immutable storage, restricted access
System Config	High	Encryption, restricted access

15.2 Compliance Requirements

Requirement	Implementation
GDPR Compliance	Data protection, right to be forgotten
Data Retention	7-year retention policy
Right to be Forgotten	Data deletion capability
Data Portability	Export user data functionality
Access Logs	All data access logged
Security Audits	Regular security assessments
Data Breach Notification	72-hour notification process

16. Testing Strategy

16.1 Test Types

Test Type	Description	Tools
Unit Tests	Test individual components	Jest, Mocha
Integration Tests	Test API endpoints	Supertest, Postman
E2E Tests	Full user workflows	Cypress, Playwright
Performance Tests	Load and stress testing	K6, JMeter
Security Tests	Vulnerability scanning	OWASP ZAP, Snyk
Accessibility Tests	WCAG compliance	Axe, Lighthouse

16.2 Testing Coverage Targets

Test Type	Coverage Target
Unit Tests	> 80%
Integration Tests	> 90% of API endpoints
E2E Tests	All critical user flows
Performance Tests	> 500 concurrent users
Security Tests	Zero critical vulnerabilities
Accessibility Tests	WCAG 2.1 AA compliance

17. Maintenance & Support

17.1 Support Tiers

Tier	Response Time	Resolution Time	Incidents
Critical	< 1 hour	< 4 hours	System down, data loss
High	< 4 hours	< 24 hours	Major feature broken
Medium	< 8 hours	< 48 hours	Minor feature issues
Low	< 24 hours	< 5 days	Enhancement requests

17.2 Maintenance Schedule

Frequency	Activities
Daily	Backup verification, log rotation
Weekly	Security patches, dependency updates
Monthly	Performance optimization, analytics review
Quarterly	Security audit, penetration testing
Yearly	Infrastructure review, upgrade planning

18. Future Enhancements

18.1 Phase 2 Features

Feature	Description	Timeline
AI-Powered Insights	Visit pattern analysis, anomaly detection	Quarter 2
Predictive Scheduling	AI-driven visit scheduling and routing	Quarter 2
Advanced Analytics	Predictive coverage models, performance scoring	Quarter 3

CRM Integration	Integration with existing CRM systems	Quarter 3
Mobile Native	QR scanning, NFC check-in, AR navigation	Quarter 4
Voice-to-Text	Voice-based report generation	Quarter 4

18.2 Scalability Roadmap

Phase	Users	Visits/Month	Infrastructure
Phase 1 (MVP)	100	1,000	Standard setup
Phase 2	500	5,000	Increased resources
Phase 3	2,000	20,000	Auto-scaling
Phase 4	10,000	100,000	Multi-region deployment

19. Glossary

Term	Definition
TC	Territory Coordinator - primary field visitor
Champion	Supervisor who monitors TCs and agents
DDIN	[Organization name - Digital financial services provider]
PWA	Progressive Web Application - web app that works offline
PostGIS	PostgreSQL extension for geographic data
JWT	JSON Web Token - for authentication

Geofence	Virtual boundary around agent location
IndexedDB	Browser-based NoSQL storage
Service Worker	Background script for offline functionality
Key Performance Indicator (KPI)	Measurable value showing system effectiveness

20. Appendices

Appendix A: Visit Record Fields

Field	Description	Source
visit_id	Unique ID for the visit	System
agent_id	Which agent was visited	System
tc_id	Which TC conducted the visit	Logged-in user
territory_id	Territory at visit time	System
visit_date	Date of the visit	System
check_in_time	When the visit started	System
check_out_time	When the visit finished	System
agent_latitude	Agent's registered location	Agent record
agent_longitude	Agent's registered location	Agent record
tc_latitude	TC's location during visit	Device GPS

tc_longitude	TC's location during visit	Device GPS
distance_meters	Distance between TC and agent	Backend calcula
gps_accuracy_meters	GPS reading accuracy	Device GPS
location_status	VERIFIED / OUTSIDE_RADIUS / LOW_ACCURACY	System
photo_id	The required visit photo	System
photo_captured_at	When the photo was taken	System
visit_status	STARTED / COMPLETED / FLAGGED / REJECTED	System
report_data	TC's structured answers	TC
remarks	Any extra notes	TC
created_at	Record creation timestamp	Server
updated_at	Record update timestamp	Server

Appendix B: Organizational Structure

text

District → Territory Coordinator (TC) → Sector → Champion → Agent

Level	Role	Coverage	Access
District	Territory Coordinator	One or more districts	Assigned districts, sectors, Champions, agents, visits
Sector	Champion	One sector	Agents and visits within assigned sector

Agent	Agent	Registered agent location	Own profile and visit-related information

Document Control

Version	Date	Author	Changes
1.0	**Sempter 2026**	**Alfred Mahame**	**Initial version**